

Chain-of-Thought Faithfulness With Ground Truth

504 agent trajectories where the correct answer is known independently of the model, and a headline result that does not survive de-duplication

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Abstract

Agentic harnesses for vulnerability localisation emit long reasoning traces and then reduce them to a ranked list of files, discarding the reasoning. Whether that discarded text contains information the structured output does not is a chain-of-thought faithfulness question, and unusually it is one with ground truth: each task is a real CVE at the commit before its fix, and the fix commit names the true files and lines.

We mine 504 trajectories for *relocations* — statements where a run contradicts the question it was given and names a different file — and report that our own headline result does not survive de-duplication. Counted per mention, relocations name a true gold file 81.4% of the time, apparently beating the 49.8% calibration of three-of-three pass agreement. Counted per distinct (task, file) claim, which is the unit the baseline is measured on, precision is 40.5% on 37 claims, below the baseline. Three tasks contribute over half the population; one is a single correct claim restated 55 times.

A separate result is unaffected: 5 of 192 relocations fail to reach the structured output, so the information is not lost in the reduction, only weighted.

We then ask whether the relocated file is recoverable *before* the model writes it, at the token where nothing on the page points at it yet. Asking the model outright recovers it in 18 of 30 distinct claims against a decoy floor of 3 ($p = 0.0003$); a Jacobian lens on the same claims recovers 11 of 29 against a floor of 5 ($p = 0.15$). The information is present and the lens does not read it. For trace mining, the unit of analysis determines the headline, and repetition within a trace is not independent evidence.

Keywords chain-of-thought faithfulness, mechanistic interpretability, LLM evaluation, agentic harnesses, code localization

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Circulated for individual review only. **Please do not redistribute, post, or cite.** Findings here are provisional: the pilot rests on ~30 distinct claims and its lens arm is not statistically significant. Two scoring rules were corrected during the work, and both the original and corrected values are reported rather than only the survivor. Numbers may change.

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1. WHY THIS SETTING HAS GROUND TRUTH

Faithfulness experiments usually cannot check a model’s stated target against an objective answer. The stated reasoning and the final output are both model artefacts; a third, independent statement of what the answer *is* rarely exists.

Vulnerability localisation supplies one. A task is a repository checked out at the parent of a known fix commit. The weakness is present at that commit and absent at its child. The fix commit is a diff, so it names the files and line ranges that were actually wrong, and it was written by the project’s own maintainers rather than by an annotator reading model output.

The model is given a symptom, the repository, and nothing else — no path, no CWE identifier, no advisory text. It explores with `glob`, `grep` and `read`, then submits files and cites lines. Its transcript, its structured output, and the true answer are three independent objects.

This yields a triple per trace: what the model **said**, what it **submitted**, and what was **true**. Faithfulness questions that are normally circular become measurable.

The corpus is 594 tasks drawn from real advisories, which produced 1,288 casefiles and 504 recorded reasoning trajectories.

2. APPARATUS

Three models with different jobs, on one workstation, with two model-free retrieval stages and a mechanical evidence stage that uses no model at all.

stage	model	job
locate	Antares-1B	ranks candidate files; run k times per task for agreement
plan / judge	Qwen3.6-35B-A3B	selects files to examine, scores
enrich	FastContext-4B	explores inside a chosen file and cites lines
evidence	—	resolves every claim against the real checkout

Everything is local, so full activations are reachable for the follow-up proposed in §7. Every run writes an ATIF trajectory (per-node step records, tool calls, reasoning text) and a casefile (the structured output, ranked candidates, cited line ranges).

Locate is run three times per task. The number of passes that name a file calibrates how likely it is to be gold: 1 of 3 arrow.r 8.0%, 2 of 3 arrow.r 22.6%, 3 of 3 arrow.r 49.8%, measured on this corpus. That calibration is the baseline any mined signal has to beat, and §4 turns on the fact that it is measured per distinct file, not per mention.

3. RELOCATIONS

A **relocation** is a statement in the reasoning where a run contradicts the question it was given and names a different file than the one it was sent to examine. It is the

simplest case where the transcript and the structured output can disagree about a checkable fact.

Most apparent contradictions are not contradictions. Four filters, validated on a held-out sample where they removed 33 of 133 raw candidates:

1. **Position within the step.** A claim made mid-reasoning and abandoned before the step ends is not the run's position.
2. **Supersession.** A claim contradicted by a later claim in the same trace is superseded, not held.
3. **Real-extension test.** Naming an additional file is an extension, not a contradiction, unless it displaces the original.
4. **Hedge detection.** "This may live in X instead" is not an assertion that it does.

192 relocations survive across the corpus, covering 37 distinct (task, file) claims.

Those two numbers are the same evidence counted two ways.

4. THE RESULT, AND ITS RETRACTION

We measured whether a relocation names a true gold file, and compared it against the three-pass agreement calibration.

unit	n	precision	vs. 49.8% baseline
per mention	192	81.4%	above
per distinct (task, file) claim	37	40.5%	below

Same data, same filters, opposite conclusions. The per-mention figure was our headline. It is wrong, for a reason that is visible once the population is decomposed:

- Three tasks contribute more than half of all mentions.
- One of them is a single correct claim restated 55 times within one trace.

A model that returns to the same conclusion as it works produces many mentions of one claim. Per mention, that reads as fifty-five successes. The baseline it was being compared against is measured per distinct file, so the comparison was never like-for-like.

The correct unit is the claim, because the claim is what a downstream consumer would act on: a file is either submitted or not, however many times the reasoning mentioned it. On that unit the mined signal is **below** the agreement baseline, and does not justify being used as a filter.

This generalises past this harness. Any pipeline that mines assertions from long transcripts can count mentions, and mentions are the default unit because they are what a regex returns. The inflated number is the one that looks publishable.

5. WHAT SURVIVED

The extraction is sound, and one finding is unaffected by the counting error because it is a question about presence rather than precision:

Of 192 relocations mined from reasoning, 5 fail to appear in the structured output.

97.4% of what the model states in its reasoning does reach the ranked list. The reduction from transcript to structured output is therefore not discarding the model's stated conclusions. This is a negative result for the "reasoning is thrown away" hypothesis in this setting.

Whether those conclusions are *weighted* correctly is a separate question. On current evidence we do not know, and §4 is the reason we decline to claim otherwise.

6. INSTRUMENT VALIDATION

Claims about a model are only worth their instrument. Two mechanisms carry that weight here.

Every claim is resolved against the real checkout on the way into a casefile. A path that does not exist at that commit is struck through, scored `p_gold 0.0`, and labelled `fabricated` rather than `dropped`, so the failure stays visible. This fires on 3 of 20 tasks in the current bank — fabricated paths are built out of the question's own vocabulary, which makes them the most plausible-looking rows in the table.

Measurements were audited against their own inputs. Eight were wrong before they were checked. Each was found by asking a number what it was made of:

reported	actual
Parser read 1,288 / 1,288 casefiles	Candidates found in 437; 831 used a five-column table the positional regex did not match
Enrichment is a no-op across four interventions	A dropped range in aggregation; the intervention was never measured
Model served with reasoning disabled	Three flags asserted it; 107 of 107 streamed tokens were reasoning
Arms differ by enricher	Locate was not replayed, so the arms received different file lists

The last two are recent. The third is instructive: a server flag named for the desired behaviour, accepted without error, that did not produce it — the request-level parameter overrode it. A flag named for the thing you want is not evidence the thing happened.

Arms are now run against a replayed locate stage, so the file list is identical across arms and the enricher is the only variable.

7. PILOT: IS THE RELOCATED FILE DECODABLE BEFORE IT IS STATED?

The follow-up below is no longer hypothetical. Two methods were put to the same question on qwen3.6-35b, the model in this corpus that both relocates often (1.10 per trace, four times any other) and has a published Jacobian lens.

Each case replays a real trajectory to the moment **before** the model names the relocated file. Two cut points test different strengths of claim: `at_path` stops mid-sentence, after `vulnerability is present in`; `at_sentence` stops before the clause exists at all. Every readout is scored a second time against a **decoy** — another case’s target, so the filename distribution matches and the answer is wrong.

Only the cut that is genuinely earlier counts. `prefix_at_sentence` falls back to the whole step when a claim has no earlier sentence boundary, and for a minority of claims it is byte-identical to `prefix_at_path` — where the model has already written in ```. Measuring “before the sentence” on those is measuring after it. Every figure below is reported on the genuinely-earlier subset, and the difference this makes is large: on the first 30-case export it moved the lens from 9/17 ($p = 0.109$) to 4/12 ($p = 0.6875$).

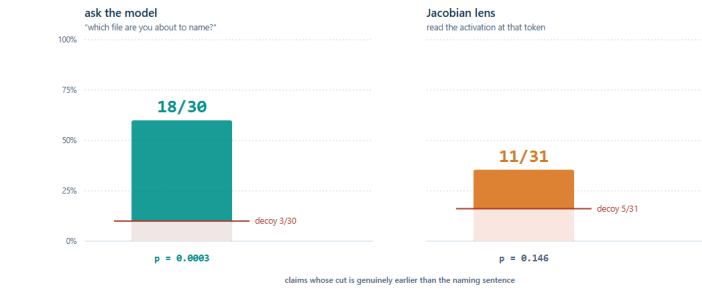
7.a. The cheap baseline beats the lens

genuinely earlier cut	claims	target	decoy	p (McNemar exact)
ask the model outright	30	18	3	0.0003
Jacobian lens, any-token scoring	31	11	5	0.146
Jacobian lens, distinctive tokens	31	11	5	0.146

The lens row is given under both scoring rules because the obvious objection is that it loses only because the permissive rule inflates its decoy. It does not: at this cut the two rules give the same 11 of 31 against the same floor of 5. The shared-token contamination is real but lives entirely at the later cut, where correcting it drops the decoy from 21 of 31 to 4.

Asking beats the lens

Same question, same corpus, same decoys: can the file be named before it is written?



The information is there and the lens does not read it. That is a claim about the instrument, not the model.
 The comparison is deliberately unfair to the lens: it reads one activation, while the baseline lets the model generate up to 2,400 tokens before answering.
 A baseline that strong losing would have settled the matter for the lens; a baseline that strong winning only bounds what the lens failed to find.

Figure 1: The two methods with their decoy floors drawn across the bars rather than beside them, because a hit rate here is unreadable without the floor it has to clear. Scored only on claims whose cut is genuinely earlier than the naming sentence.

Asking is simply appending “*which file are you about to name? Reply with one path*” to the same prefix and reading the reply. It recovers the file in 18 of 30 claims against a decoy floor of 3. The lens, on the same question and the same corpus, does not separate from its decoy.

The information is there. The lens does not read it. That is the pilot’s result, and it is a statement about the instrument rather than about the model. It also makes the faithfulness question well-posed: something at that point in the trajectory determines the file the model is about to name, and it is recoverable — so “can a lens recover it?” has a real answer, and for this lens the answer is no.

The comparison is deliberately unfair to the lens, in the direction that makes the negative safe: the lens reads a single activation, while the baseline lets the model generate — up to 2,400 tokens, and on these cases it uses them. So the baseline shows the answer is reachable *with additional forward computation*, not that it sits in the residual stream. A baseline that strong losing would have been decisive for the lens; a baseline that strong winning only bounds what the lens failed to find.

Where the file becomes decodable

Fraction of 17 claims with the target in the top 20, by layer, distinctive tokens only



The transport reads the relocated file out of the middle of the stack; the plain logit lens reaches it only at layer 38, where it is decoding the answer the model is about to say. That is the case for the transport doing real work. It is NOT a case for the strong claim: at_sentence peaks at 29% of claims and stays at $p = 0.109$, unchanged by distinctive-token scoring. Both panels are the same 17 claims, scored identically; only the cut point differs.

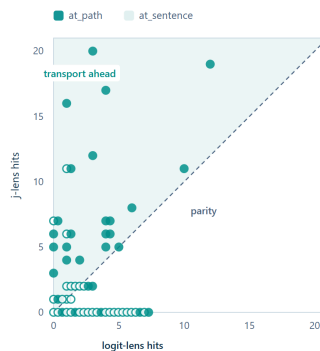
Figure 2: Where the relocated file becomes decodable, by layer, on the first 30-case export. At **at_path** the transport recovers the target from layer 20 and reaches 53% of claims by layer 30; the plain logit lens sits at or below 6% through layer 34 and arrives only at layer 38 — the final layer, where it is decoding the answer the model is about to emit. The decoy stays flat below 6%. This figure describes the later cut, where the model is already mid-sentence; it is not evidence about the earlier cut, where the lens does not beat its decoy.

7.b. What survives about the transport

Against the plain logit lens the Jacobian transport is **never worse on any of the 60 readouts** of the first export, carrying $4.2\times$ the layer-hits before the naming sentence and $2.7\times$ after. That is a real property of the transport and it is orthogonal to the result above: being a better readout than the logit lens does not make it a sufficient one.

The transport is never worse

j-lens against plain logit lens, per readout, same forward pass

**0 of 60**

rows where the transport does worse

4.2x

layer-hits before the sentence exists (38 against 9)

2.7x

layer-hits once it has begun (195 against 72)

16 vs 14

per CLAIM at at_path — the binary hides the margin

Per claim the two lenses look close at at_path, and the report says so. Per readout they are not close: the transport leads on 24 of 30 rows there, 10 of 30 before the sentence, and trails on none. The claim-level test is the conservative one and stays the headline.

Figure 3: Jacobian transport against the plain logit lens, per readout, same forward pass. Every point sits on or above the diagonal. Per *claim* the two look close at

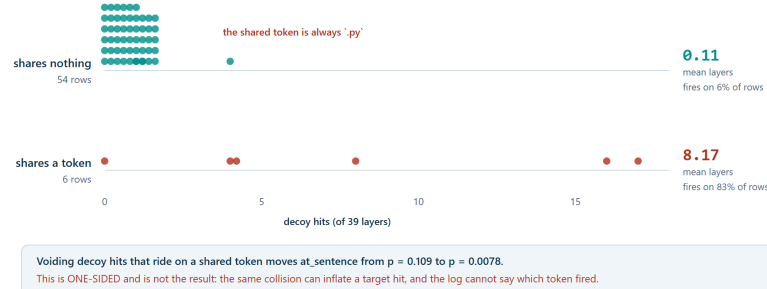
at_path — 16 of 17 against 14 of 17 — which is the conservative test and stays the headline.

7.c. The scoring defect the control found

A hit was counted when **any** token of a filename entered the top 20. Two unrelated files sharing .py therefore scored for the target and the decoy at once.

The decoy is silent unless it shares a token

60 readouts, split by whether decoy and target have any token in common



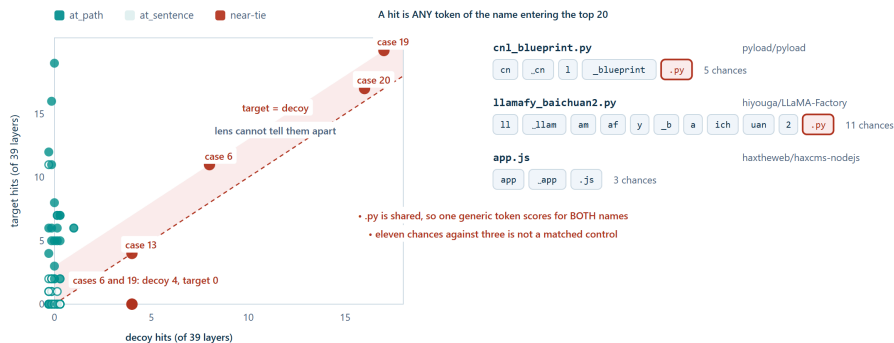
Name LENGTH is not the confound — Pearson r between token count and hits is 0.183 over 120 observations, and the 4-token name outscores the 11-token one. Shared tokens are. The fix is to score on distinctive tokens only, symmetrically, which needs the re-run rather than a re-analysis of this log.

Figure 4: The decoy's noise floor, split by whether it shares a token with the target. Sharing nothing it fires on 6% of readouts at 0.11 mean layers; sharing one token — always .py — on 83% at 8.17. Name *length* was the first hypothesis and the data rejected it: $r = 0.183$ over 120 observations, and the four-token name outscores the eleven-token one.

Correcting it symmetrically — dropping shared tokens from target and decoy alike — tightened at_path and left at_sentence unchanged. We had expected it to rescue the result; a one-sided correction appeared to, and was not licensed.

What the decoy caught

60 readouts, each scored against the target and against another case's target



Five of the six near-ties are cn1_blueprint.py against llamafy_baichuan2.py, in both directions — unrelated files in unrelated repos. The control is doing its job: it flags the rows where a hit is a shared subword rather than the file. Scoring on distinctive tokens only is the fix, and it needs a re-run, because the probe recorded how many layers hit and never which tokens they hit on.

Figure 5: Why the decoy is not decoration. Most readouts sit against the vertical axis. The six in the tie band are traceable to a two-character token.

7.d. Provenance of the numbers

Two exports and two runs sit behind this section, and they measure different things:

	cases	claims	cut taken from	raw readout
first export	30	17	message	complete — top-20 ids at 39 layers
second export	75	34	reasoning_content	complete — 149 readouts, all 75 cases

The first export only found claims whose path also appeared in the visible message, so its cut sat in the wrong text. Its figures are retained above only where they describe the transport or the scoring defect — neither of which depends on the cut being early.

The second export reads the field the miner actually found the claim in and is $2.5\times$ larger; it is the one the headline table is computed on. Its lens run was pruned by the compute provider at row 145 of 150, but the probe appends each readout as it is produced rather than writing at the end, so the rows survived; the two remaining cases were scored by a short resume run and merged. The resume reproduces the seeded decoy pairing exactly, because the probe iterates the whole case list and skips rather than slicing it — slicing would reseed the shuffle and re-pair every target.

8. PROPOSED FOLLOW-UP

The divergence between stated and submitted location is visible in the transcript, after the fact. The question this dataset is positioned to ask is whether it is visible **in the activations, and before the output is produced**. The pilot above is a first pass at question 1; questions 2 and 3 are untouched.

Concretely, the corpus supplies per trace a triple — stated location, submitted location, true location — with 192 cases where stated and submitted diverge and 5 where a stated location never reaches the output at all. The models are 1B–4B and run locally, so residual streams are accessible without inference-provider constraints.

Three questions, in increasing order of interest:

1. **Is the submitted answer decodable before it is stated?** Train a linear probe on the residual stream at the token where the model states location A, predicting the file it will eventually submit. If the eventual answer is already linearly decodable, the stated location is not the computation that produced it.
2. **Do divergent traces look different at the point of divergence?** Compare probe confidence and representational geometry between traces where stated and submitted agree and the 192 where they do not, at matched positions.
3. **Is the 5-case set — stated but never submitted — mechanistically distinct** from the cases where a stated location is submitted and wrong? One is a suppression story, the other an error story, and they should not look alike.

The value of the setting is that (1) can be scored against an answer key that neither the transcript nor the output produced.

9. LIMITATIONS

- 37 distinct claims is a small denominator. The per-claim precision of 40.5% has a wide interval, and the honest reading is “below baseline, not usefully estimated”.
- Three tasks dominating the mention population means the corpus is not diverse at the trace level, whatever the task count suggests.
- Relocation is one operationalisation of stated/acted divergence and the narrowest one. It does not capture a run that reasons correctly and submits correctly for a stated reason that is false.
- The agreement calibration (49.8% at 3 of 3) is measured on this corpus and this specialist. It is a local baseline, not a general one.
- Gold ranges come from fix-commit hunks with context, so line-level scoring inherits whatever the maintainer included in the diff.
- The pilot has ~30 distinct claims from 15 repositories. That is enough to establish the baseline result ($p = 0.0003$) and not enough to distinguish a weak lens effect from none.
- The lens arm covers all 75 cases. The main run was pruned by the provider at row 145 of 150; the rows survived because the probe appends each readout as it is produced rather than writing at the end, and the two remaining cases were scored by a short resume run and merged. The decoy pairing is identical across both because the probe iterates the full case list and skips rather than slicing it.
- The baseline lets the model generate up to 2,400 tokens before answering, while the lens reads one activation. The comparison is deliberately unfair in the direction that makes the lens negative safe, and correspondingly weak as evidence that nothing is in the residual stream.
- Lens scoring counts a hit when any token of a filename enters the top 20, which is permissive. Shared tokens are now removed from target and decoy alike, which fixes the collision the control exposed, but the rule is still “any token at any layer” and a hit is not a rank.
- The emergence curve is a per-layer *hit rate over claims*, not a confidence trajectory. It says where the token enters a top-20, not how strongly. Layer 38 is the model’s own output distribution, so the logit lens arriving there is expected and is not evidence about the transport.